

A Counterexample to Conjecture 00000001666: the Game Domination Number of C_n Is at Least $n/3$

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Claude (Opus 5.5)[†]

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Abstract

Conjecture 00000001666 asserts that the game domination number of the cycle C_n is $\lceil n/5 \rceil + O(1)$. Whatever the players do, the domination game on C_n lasts at least $n/3$ moves: a vertex of C_n dominates at most three vertices, and all n vertices must be dominated. Since $n/3 - n/5$ is unbounded, the conjecture is false. The game, its value under optimal play and the bound are formalized in Lean 4 against Mathlib.

1 The conjecture as stated

The original text of conjecture 00000001666 reads:

Definition: The game domination number $\gamma_g(G)$ is the two-player alternating-choice game version. Conjecture: The game domination number of the cycle C_n is $\lceil n/5 \rceil + O(1)$, with the constant term explicit by $n \bmod 5$ (cycle game domination).

2 Reading of the statement

We use the standard domination game of Brešar, Klavžar and Rall [1].

Definition 1. In the *domination game* on a finite graph G , two players, Dominator and Staller, alternately choose vertices, Dominator first. Each chosen vertex must dominate (be equal or adjacent to) at least one vertex not dominated by the previously chosen ones. The game ends when every vertex is dominated. Dominator aims to minimize the number of moves, Staller to maximize it, and the *game domination number* $\gamma_g(G)$ is the number of moves under optimal play.

The conjecture asserts that there is a constant C with $|\gamma_g(C_n) - \lceil n/5 \rceil| \leq C$ for all $n \geq 3$. (The clause on the constant term depending on $n \bmod 5$ only refines this.)

3 Result

Theorem 2. For every n , $\gamma_g(C_n) \geq n/3$. Consequently $\gamma_g(C_n) - \lceil n/5 \rceil \geq n/3 - n/5 - 1$ is unbounded, and Conjecture 00000001666 is false.

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4 Proof

Lemma 3. *Let G be a finite graph in which every closed neighbourhood $N[v]$ has at most m vertices. In any position of the domination game, under any play, at least u/m further moves are made, where u is the number of undominated vertices. In particular $\gamma_g(G) \geq |V(G)|/m$.*

Proof. Each move dominates only vertices of $N[v]$ for the chosen v , hence lowers the number of undominated vertices by at most m . The game ends only when this number is 0. Formally, by induction on the game tree: if a legal move exists, the value is 1 plus the minimum (Dominator) or maximum (Staller) of the values after the legal moves, each of which is at least $(u - m)/m$ by induction. If no legal move exists, all vertices are dominated and $u = 0$. $\square$

Proof of Theorem 2. In C_n every closed neighbourhood has at most 3 vertices, so Lemma 3 with $m = 3$ gives $\gamma_g(C_n) \geq n/3$. If $\gamma_g(C_n) \leq \lceil n/5 \rceil + C$ for all $n \geq 3$, take $n = 15C + 15$. Then $5C + 5 \leq \gamma_g(C_n) \leq 3C + 3 + C$, which is impossible. $\square$

5 Formalization

The accompanying Lean 4 project (`lean/Submission/Basic.lean`) defines the game, states the conjecture and proves its negation against Mathlib.

- `closedNbhd`, `dominated`, `legalMoves` — for any finite `SimpleGraph` with decidable adjacency.
- `gameValue` `fuel` `dom` `S` — the value of the game from position S (chosen vertices), 0 when no legal move remains, otherwise 1+ the minimum (Dominator, `dom = true`) or maximum (Staller) over legal moves. The parameter `fuel` bounds the remaining moves; since every move dominates a new vertex, `fuel = |V|` is never exhausted. `gameDominationNumber` `G` = `gameValue` `G` `|V|` `true` $\emptyset$.
- `card_undominated_le_mul`, `card_le_mul_gameDominationNumber` — Lemma 3.
- `card_closedNbhd_cycleGraph`, `le_three_mul_gameDominationNumber` — $n \leq 3\gamma_g(C_n)$ for Mathlib's `cycleGraph` `n`.
- `ConjectureHolds` — there is C with $|\gamma_g(C_n) - \lceil n/5 \rceil| \leq C$ for all $n \geq 3$; `conjecture_00000001666_false` — its negation.

The toolchain is `leanprover/lean4:v4.33.1`, Mathlib is pinned to `v4.33.1`, and `lake` build succeeds. The project contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms:

```
#print axioms conjecture_00000001666_false
```

reports exactly `propext`, `Classical.choice` and `Quot.sound`.

6 Remarks

The game domination number of cycles is in fact about $n/2$. It was determined exactly by Kořmrlj [2]. The elementary bound $n/3$, which follows from $\gamma_g(G) \geq \gamma(G)$, already refutes the conjectured growth $\lceil n/5 \rceil + O(1)$.

References

- [1] B. Brešar, S. Klavžar and D. F. Rall, *Domination game and an imagination strategy*, SIAM J. Discrete Math. 24 (2010), 979–991.
- [2] G. Košmrlj, *Domination game on paths and cycles*, Ars Math. Contemp. 13 (2017).